Introduction

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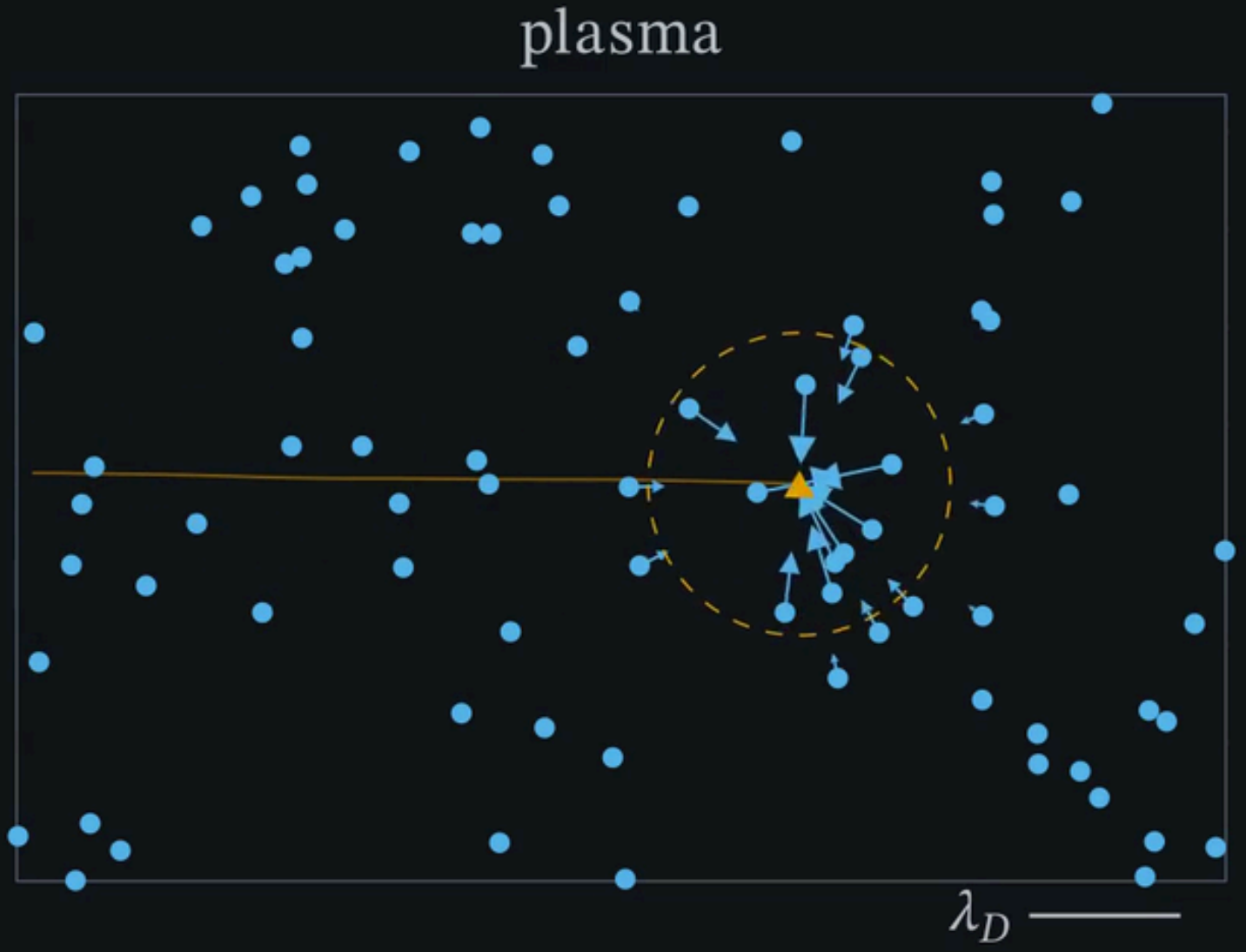
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plasma
λ_D

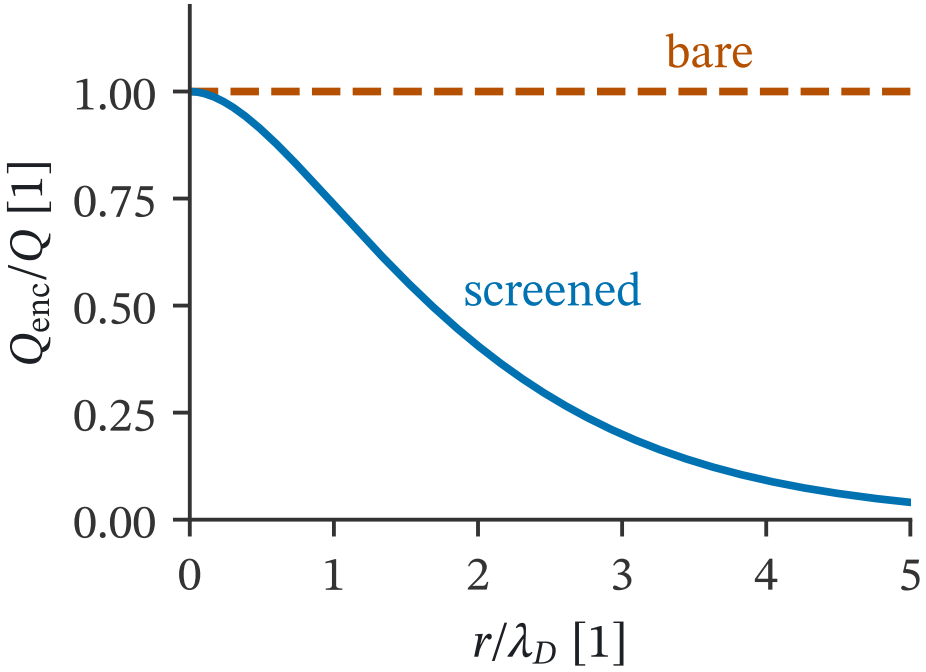
1.1 Quasineutrality

$$\rho_q = e\left(\sum_i Z_i n_i - n_e\right)$$

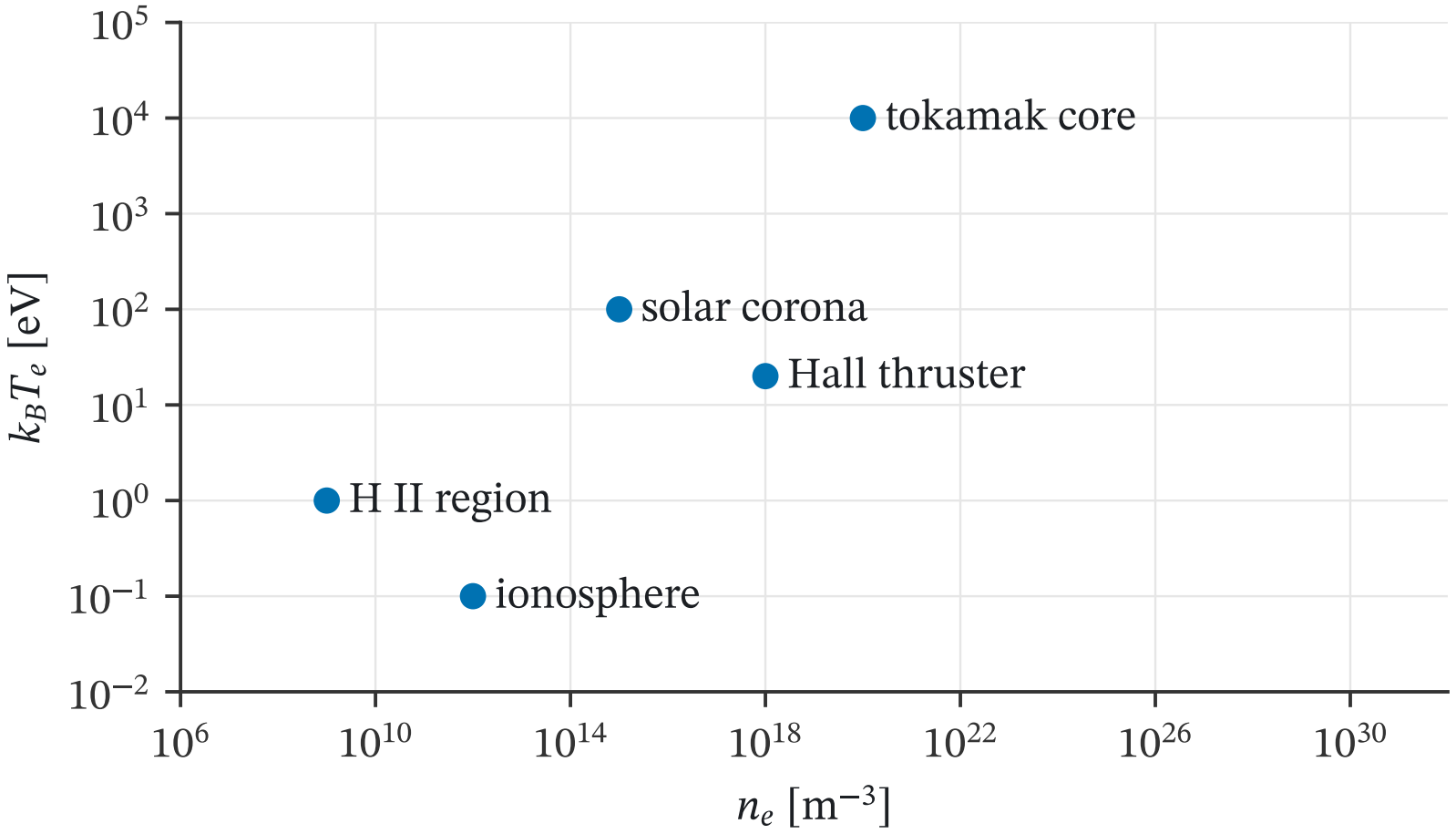
$$n_e \approx \sum_i Z_i n_i$$

$L \gg \lambda_D$

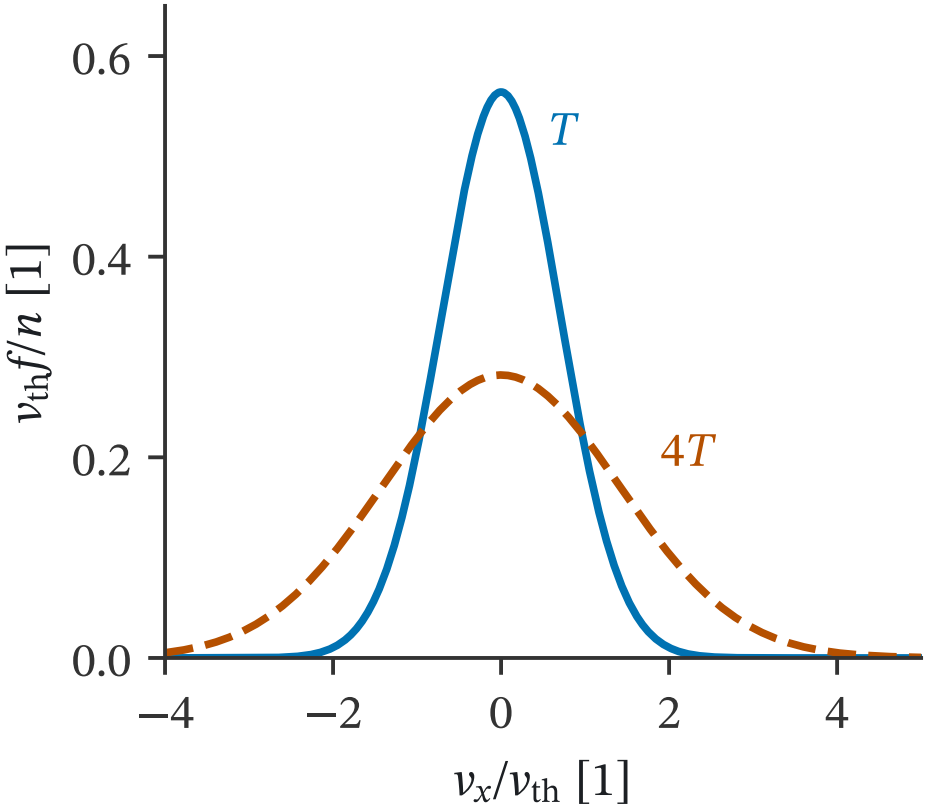
L observation length
λ_D screening length (chapter 4)

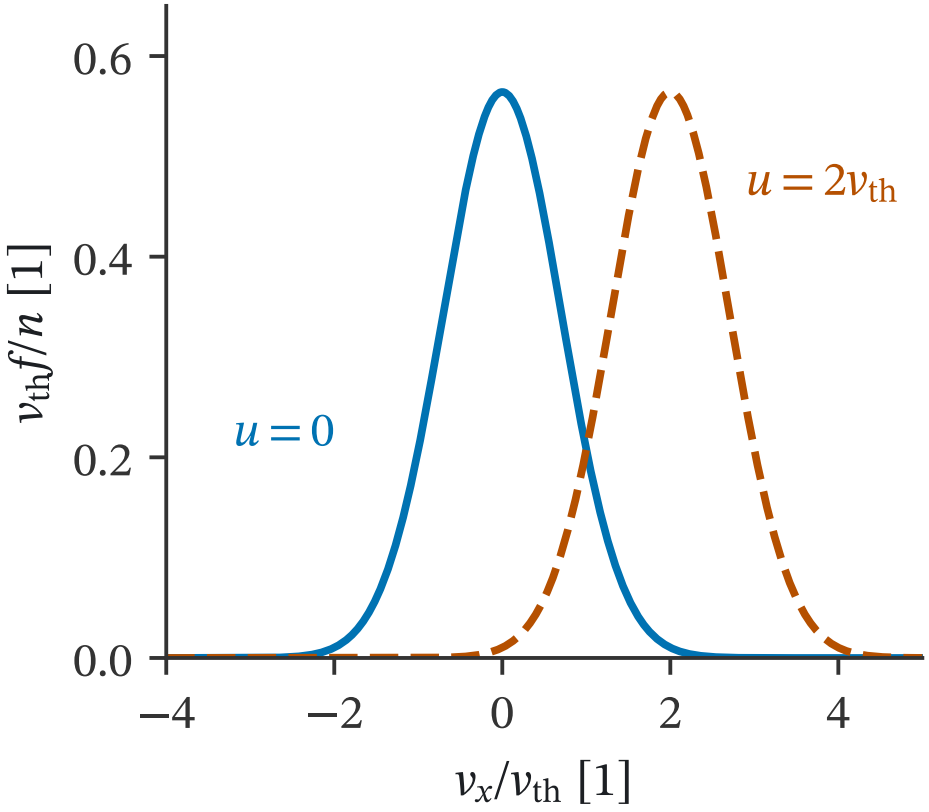


1.1 Example plasmas



1.2 Thermal motion





1.2 Thermal speed

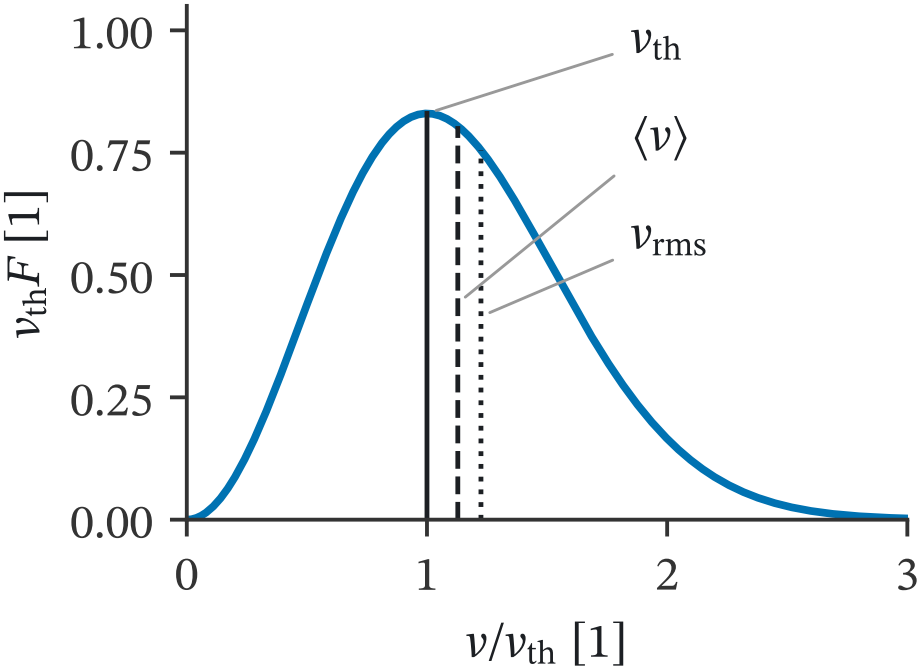
$$v_{\mathrm{th},s} = \sqrt{\frac{2k_B T_s}{m_s}}$$

thermal-speed convention

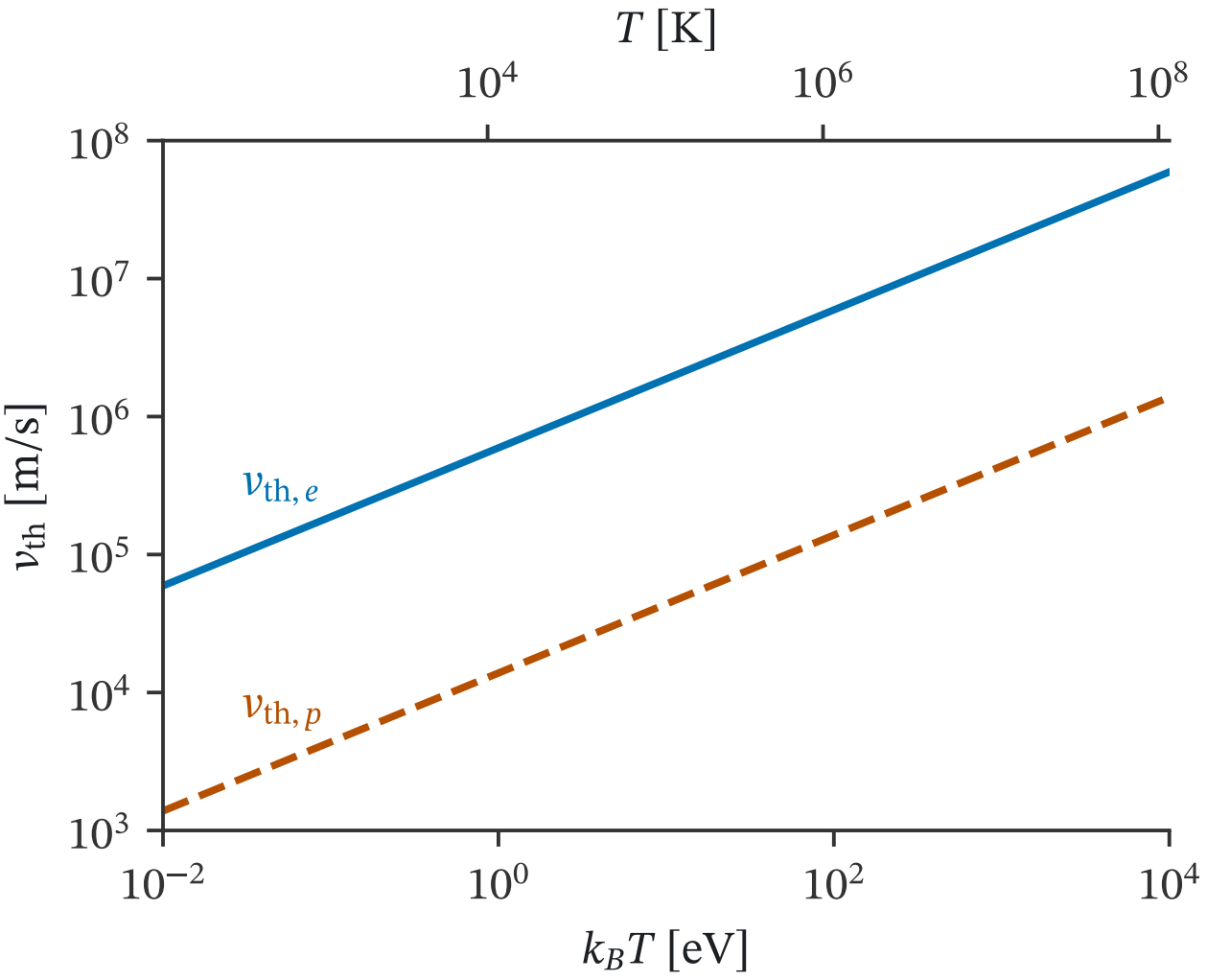
$$\langle (v_x - u_s)^2 \rangle = \frac{k_B T_s}{m_s}$$

$$v_{\mathrm{peak}} = v_{\mathrm{th},s}, \quad v_{\mathrm{rms}} = \sqrt{\frac{3}{2}} v_{\mathrm{th},s}$$

m_s mass; u_s mean flow
T_s temperature



1.2 Temperature scales



$$k_B T = 1 \text{ eV} \quad \leftrightarrow \quad T \approx 11605 \text{ K}$$

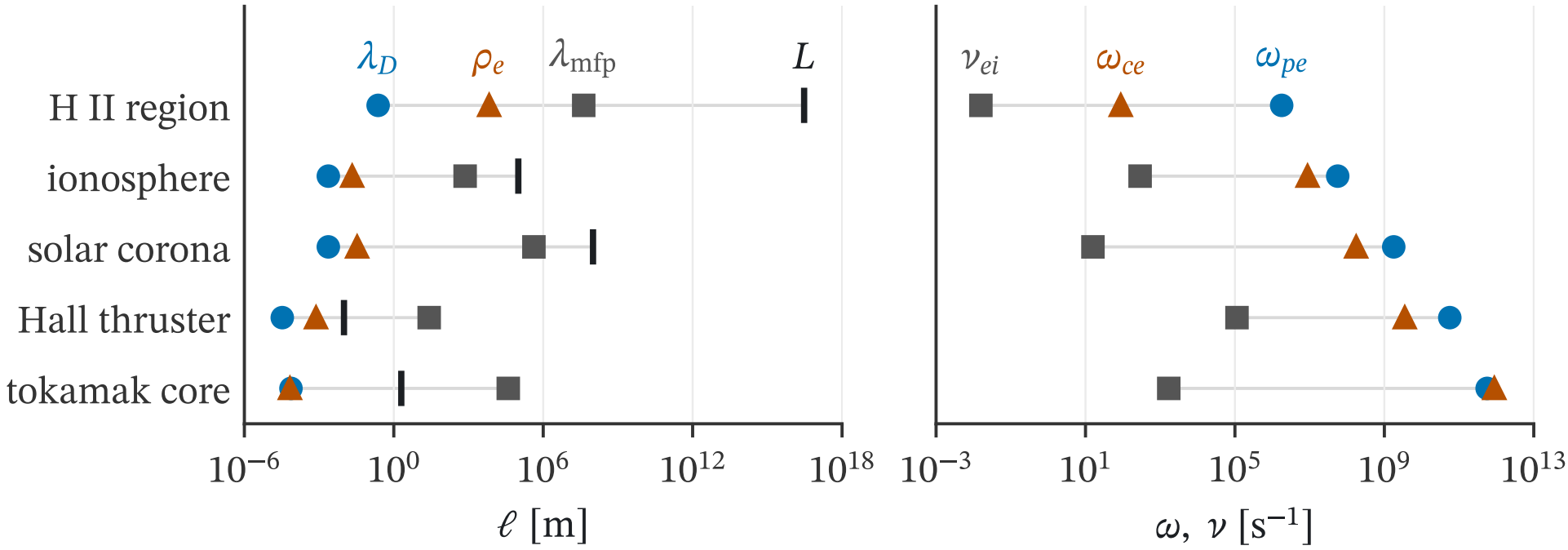
1.3 Characteristic scales

$$\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_e e^2}}$$

$$\omega_{pe} = \sqrt{\frac{n_e e^2}{\varepsilon_0 m_e}}$$

$$\rho_e = \frac{v_{\mathrm{th},e}}{\omega_{ce}}$$

$$\lambda_{\mathrm{mfp}} = \langle v \frac{\rangle}{\nu_{ei}}$$



1.3 Plasma phenomena

λ_D

screening

ω_{pe}

oscillation

$N_D \gg 1$

collective interaction

$\boldsymbol{j}, \boldsymbol{q}$

charge and heat transport

$q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$

electromagnetic response

$h\nu$

radiation

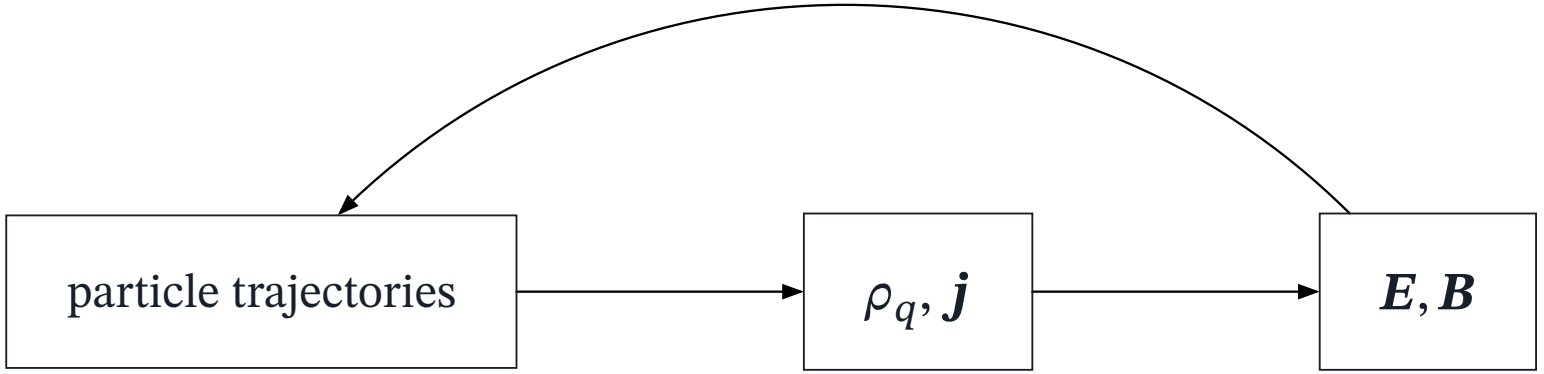
1.4 Microscopic dynamics

$$\frac{\mathrm{d}\boldsymbol{r}_j}{\mathrm{d}t} = \boldsymbol{v}_j, \quad m_j \frac{\mathrm{d}\boldsymbol{v}_j}{\mathrm{d}t} = q_j\left(\boldsymbol{E} + \boldsymbol{v}_j \times \boldsymbol{B}\right)$$

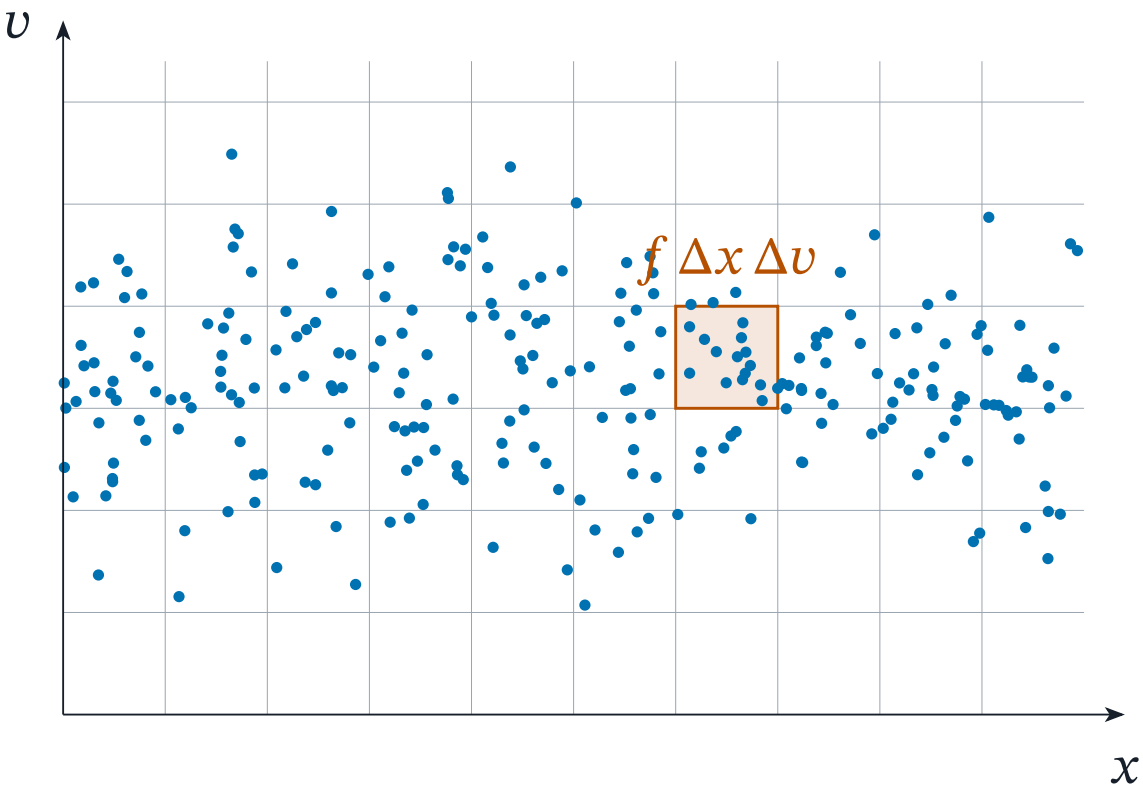
N coupled trajectories

$N \sim 10^{20}$
$\sim N^2$ pair interactions

self-consistent $\boldsymbol{E}, \boldsymbol{B}$
Maxwell field equations



1.4 Distribution function

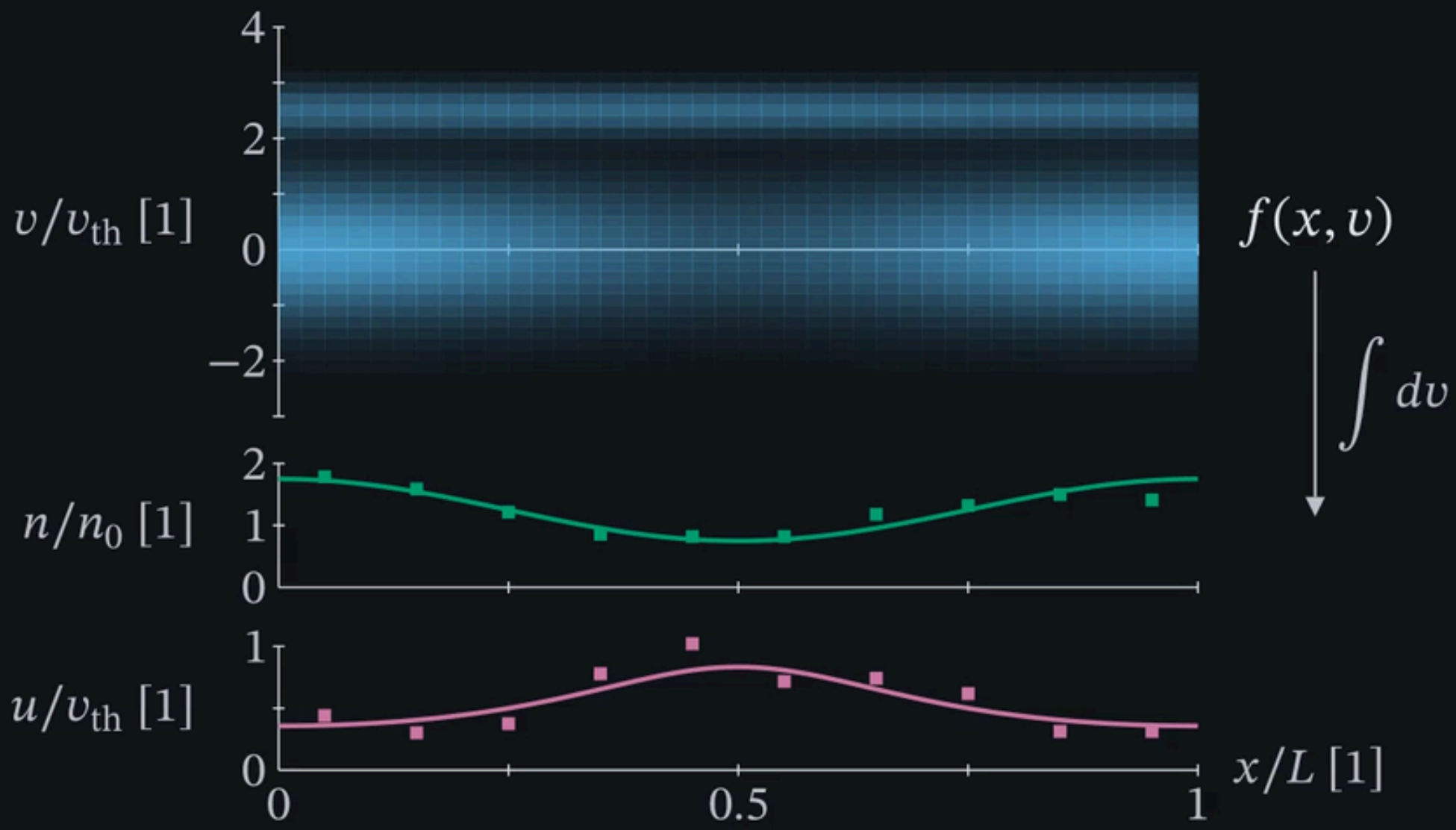


$$f_s(\boldsymbol{x}, \boldsymbol{v}, t)\mathrm{d}^3x\mathrm{d}^3v$$

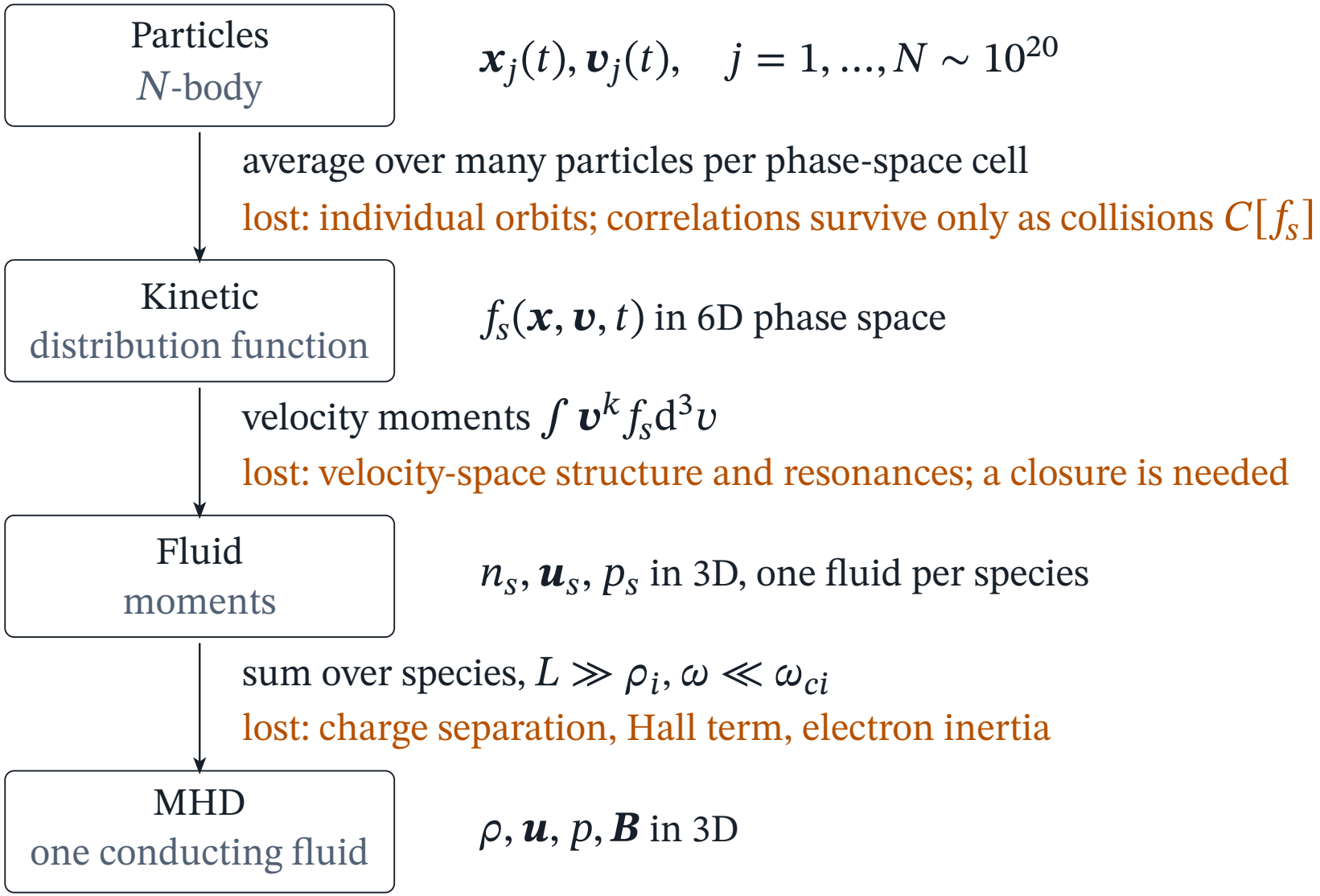
particles of species s in a phase-space cell

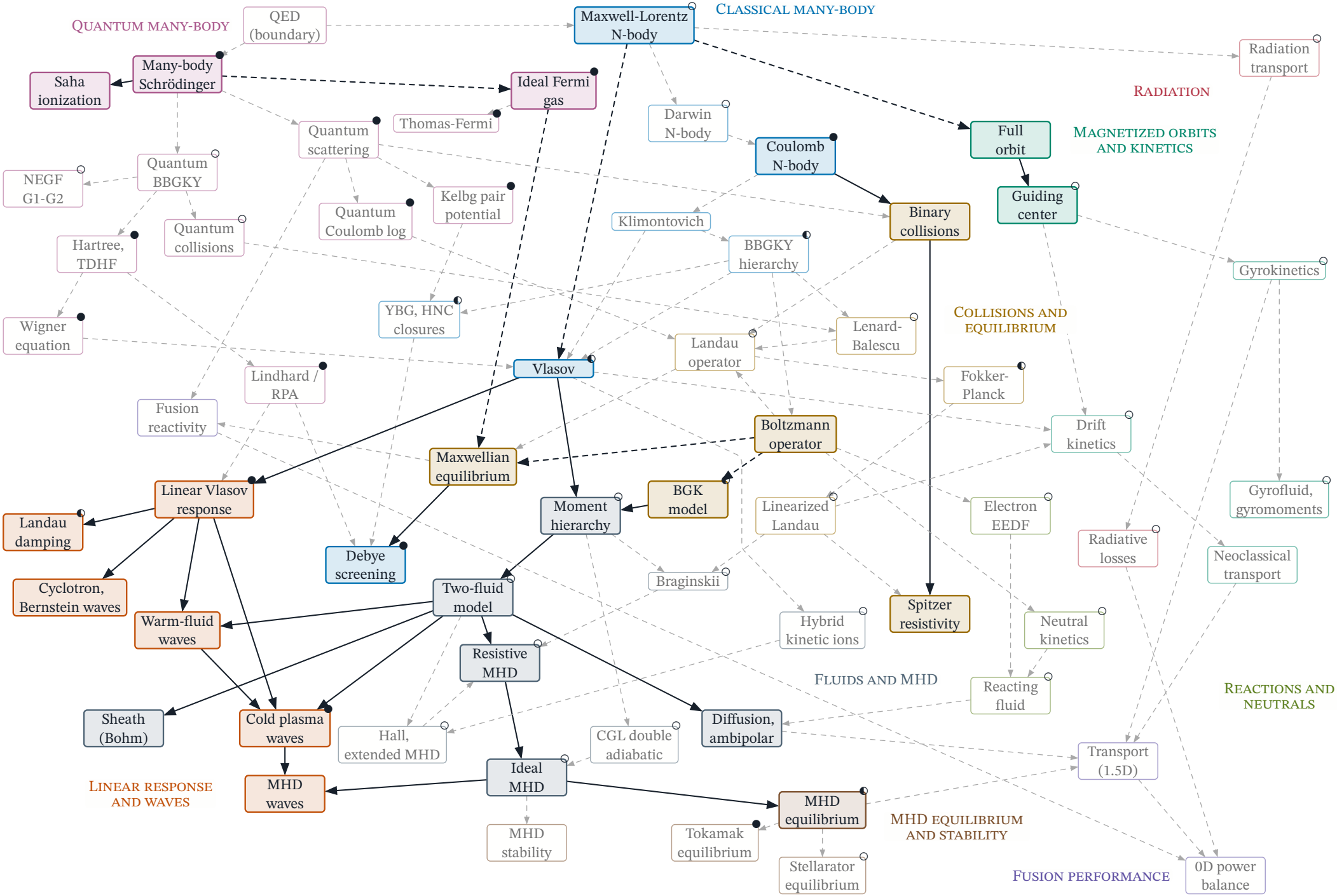
$$n_s(\boldsymbol{x}, t) = \int f_s \mathrm{d}^3v$$

f_s in $\mathrm{s}^3\mathrm{m}^{-6}$; n_s number density


4
2
0
−2
v/v_{th} [1]
$f(x, v)$
$\int dv$
2
1
0
n/n_0 [1]
1
0
u/v_{th} [1]
0
0.5
1
x/L [1]

1.4 Model hierarchy




QUANTUM MANY-BODY
QED (boundary)
Many-body Schrödinger
Saha ionization
Ideal Fermi gas
Thomas-Fermi
Quantum scattering
Quantum BBGKY
NEGF G1-G2
Quantum collisions
Quantum Coulomb log
Kelbg pair potential
Hartree, TDHF
Wigner equation
Lindhard / RPA
Fusion reactivity
YBG, HNC closures
CLASSICAL MANY-BODY
Maxwell-Lorentz N-body
Darwin N-body
Coulomb N-body
Klimontovich
BBGKY hierarchy
Vlasov
Landau operator
Lenard-Balescu
Maxwellian equilibrium
Boltzmann operator
BGK model
Moment hierarchy
Linear Vlasov response
Landau damping
Debye screening
Cyclotron, Bernstein waves
Warm-fluid waves
Two-fluid model
Braginskii
Resistive MHD
Hall, extended MHD
CGL double adiabatic
Ideal MHD
Sheath (Bohm)
Cold plasma waves
MHD waves
LINEAR RESPONSE AND WAVES
MHD stability
MHD equilibrium
Tokamak equilibrium
Stellarator equilibrium
MHD EQUILIBRIUM AND STABILITY
Diffusion, ambipolar
FLUIDS AND MHD
Hybrid kinetic ions
Spitzer resistivity
Linearized Landau
Electron EEDF
Neutral kinetics
Reacting fluid
RADIATION
Radiation transport
MAGNETIZED ORBITS AND KINETICS
Full orbit
Guiding center
Binary collisions
COLLISIONS AND EQUILIBRIUM
Fokker-Planck
Drift kinetics
Gyrokinetics
Gyrofluid, gyromoments
Radiative losses
Neoclassical transport
REACTIONS AND NEUTRALS
Transport (1.5D)
FUSION PERFORMANCE
0D power balance
checked symbolically
stated
route of this course
kin6d: implemented
partial
planned

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